

Erdős Problem #873

EP-873

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $A = \{a_1 < a_2 < \dots\} \subseteq \mathbb{N}$ and let $F(A, X, k)$ count the number of i such that $[a_i, a_{i+1}, \dots, a_{i+k-1}] < X$, where the left-hand side is the least common multiple. Is it true that, for every $\epsilon > 0$, there exists some k such that $F(A, X, k) < X^\epsilon$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>